Exponentials & Logarithms

Section 1: Exponential Equations & Radicals

1 Solve the exponential equation for x: $27^{x-1} = 9^{x+2}$.

a. 5

b. 6

c. 7

d. 8

2 Simplify the radical expression: $(81x^8y^{12})^{1/4}$ for positive x and y.

a. $3x^2y^3$

b. $9x^2y^3$

c. $3x^4y^8$

d. $9x^4y^8$

3 If $2^x = 5$, what is the value of 2^{x+3}?

a. 8

b. 15

c. 40

d. 125

Section 2: Logarithmic Properties & Evaluation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Evaluate $\log_2(64) - \log_3(27)$.

a. 1

b. 2

c. 3

d. 4

5 Condense into a single logarithm: $2\log x + 3\log y - \log z$.

a. $\log(\frac{x^2y^3}{z})$

b. $\log(\frac{2x \cdot 3y}{z})$

c. $\log(x^2 + y^3 - z)$

d. $\log(\frac{x^2z}{y^3})$

6 If $\log_{10}(2) \approx 0.301$, what is the approximate value of $\log_{10}(20)$?

a. 1.301

b. 2.301

c. 0.602

d. 3.010

Section 3: Solving Logarithmic Equations

7 Solve for x: $\log_5(2x + 7) = 2$.

a. 9

b. 11

c. 18

d. 25

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Solve for x: $\log(x) + \log(x-3) = 1$ in base 10.

a. 5

b. 2

c. -2

d. 5 and -2

9 What is the solution to $\ln(e^{3x-1}) = 8$?

a. 2

b. 3

c. 4

d. $\frac{7}{3}$

Section 4: Exponential Growth, Decay & Compound Interest

10 A radioactive isotope decays according to $N(t) = 800(0.5)^{t/12}$, where t is measured in days. How much of the isotope remains after 36 days?

a. 50 g

b. 100 g

c. 200 g

d. 400 g

11 A principal of $5,000$ is invested at an annual interest rate of 4% compounded quarterly. Which expression represents the balance after 5 years?

a. $5000(1 + \frac{0.04}{4})^{20}$

b. $5000(1 + 0.04)^{5}$

c. $5000(1 + \frac{0.04}{12})^{60}$

d. $5000e^{0.20}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 A population of fish grows by 8% each year. If the initial population is $1,200$, what is the population after t years?

a. $1200(1.08)^t$

b. $1200(0.92)^t$

c. $1200(1 + 0.8)^t$

d. $1200 + 8t$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647